



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

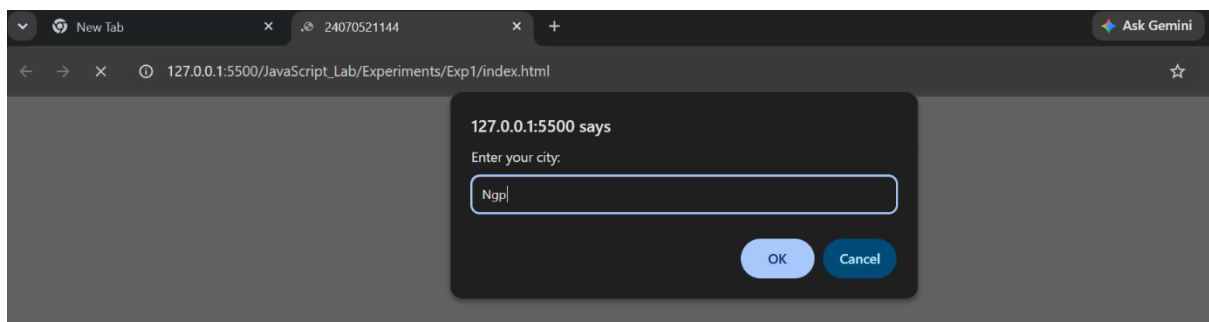
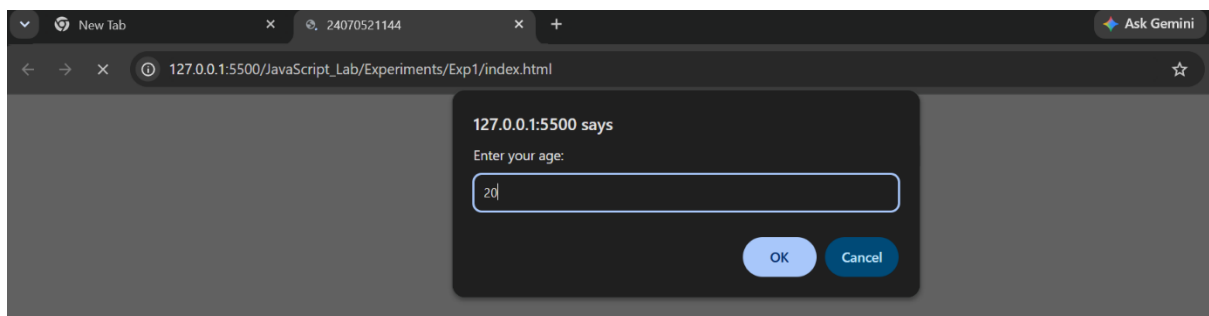
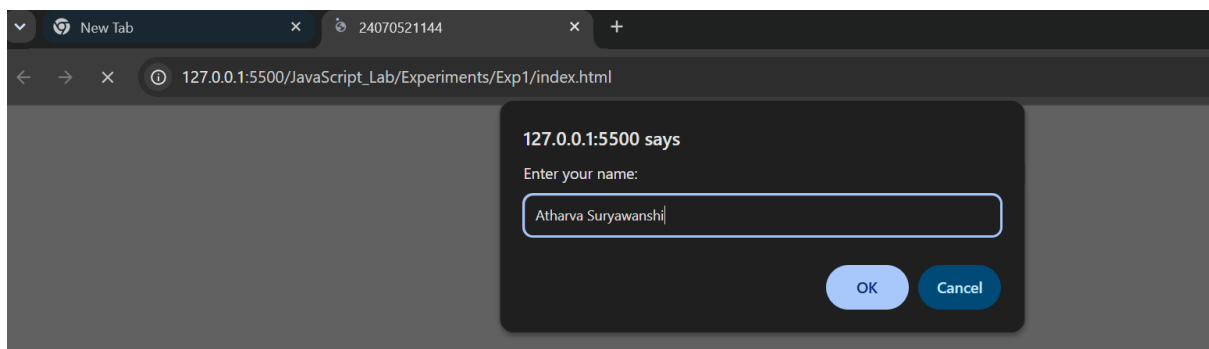
Experiment 01

Aim: Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Theory: JavaScript can be included in a web page using inline, internal, and external methods. Inline JavaScript is written directly within HTML elements, internal JavaScript is placed inside the

Output:





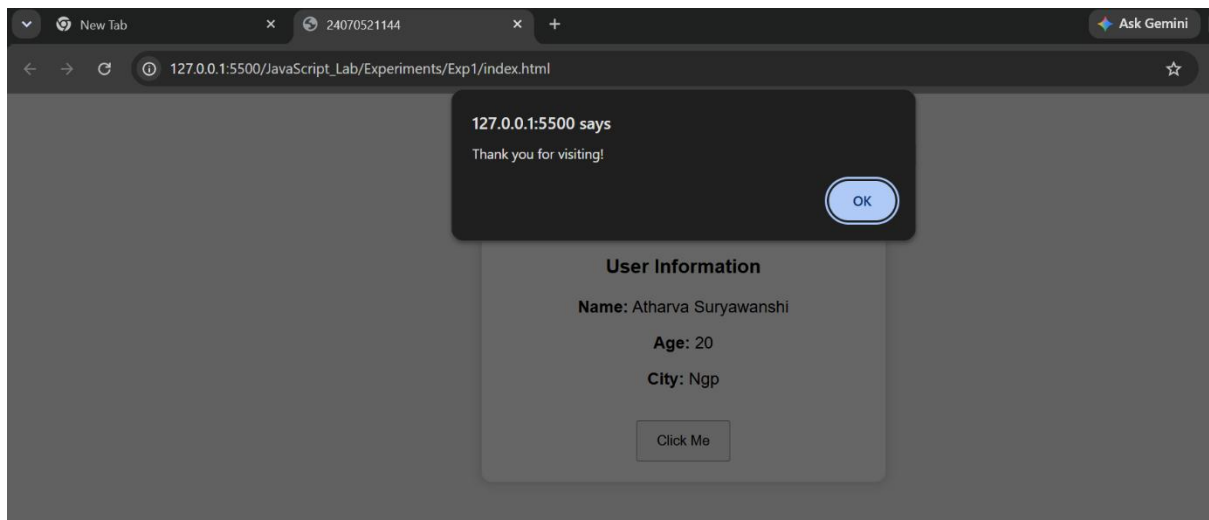
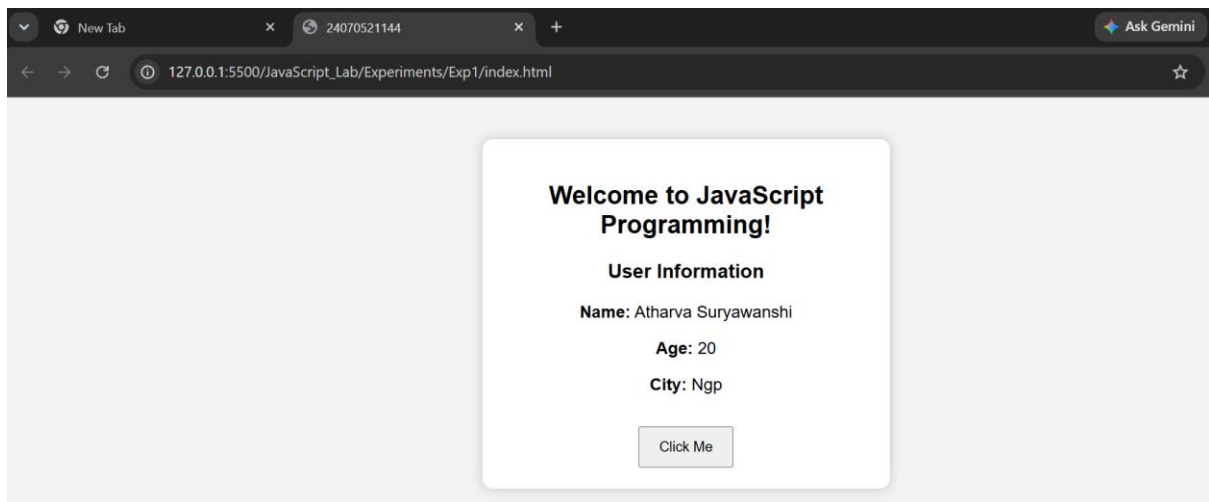
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Result: The webpage was successfully created using inline, internal, and external JavaScript. It displayed a welcome message, accepted user information through input prompts, and demonstrated various console methods such as `console.log()`, `console.info()`, `console.warn()`, `console.error()`, and `console.table()` for debugging and output.



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Case Study 01

Aim: Create a webpage for college department and link the student information page with it and write a javascript program to display “Welcome to SIT Nagpur” in an alert box.

Program Code:

```
Case_Study > Case_Study1 > student_info.html > ...
You, 2 weeks ago | 1 author (You)
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  <meta charset="UTF-8">
5  <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  <title>Student Information</title>
7  <style> ...
46 </style>
47 </head>
48 <body>
49
50 <h1>Student Information Directory</h1>
51 <p>Below is the list of current students enrolled in the department.</p>
52
53 <table>
54   <thead>
55     <tr>
56       <th>Roll Number</th>
57       <th>Name</th>
58       <th>Year</th>
59       <th>Email</th>
60     </tr>
61   </thead>
62   <tbody>
63     <tr>
64       <td>CS101</td>
65       <td>Alice Smith</td>
66       <td>3rd Year</td>
67       <td>alice@college.edu</td>
68     </tr>
69     <tr>
70       <td>CS102</td>
71       <td>Bob Jones</td>
72       <td>2nd Year</td>
73       <td>bob@college.edu</td>
74     </tr>
75     <tr>
76       <td>CS103</td>
77       <td>Charlie Brown</td>
78       <td>4th Year</td>
79       <td>charlie@college.edu</td>
80     </tr>
81   </tbody>
82 </table>
83
84 <a href="department.html" class="back-btn">Back to Department Home</a>
85
86 </body>
87 </html>
```



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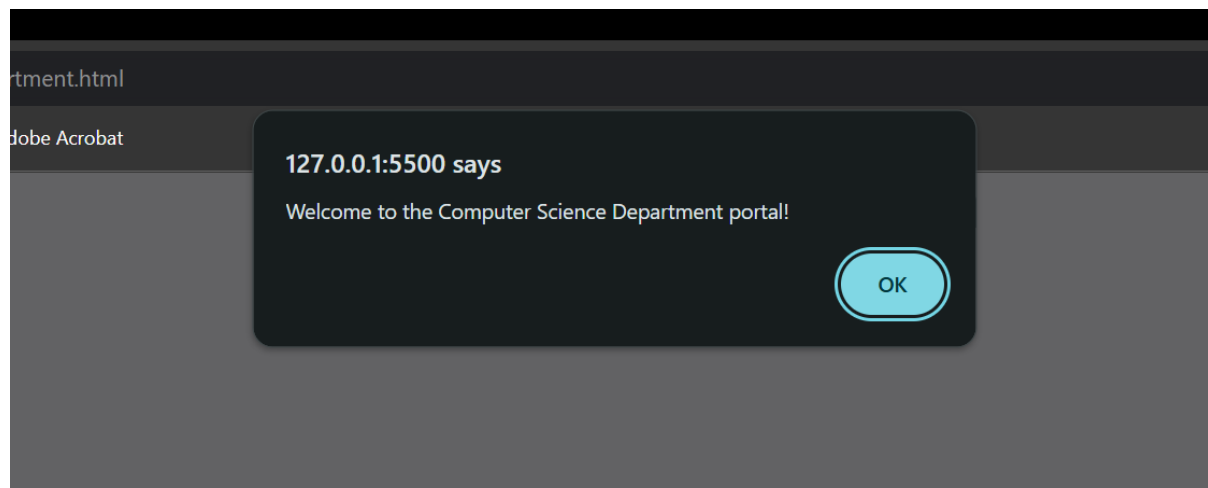
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```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
6    <title>Computer Science Department</title>
7  </head>
52 </style>
53 </head>
54 <body>
55
56 <header>
57   <h1>Symbiosis Institute of Technology</h1>
58   <p>Wathoda Nagpur</p>
59 </header>
60
61 <nav>
62   <a href="#">Home</a>
63   <a href="student_info.html">Student Information</a>
64   <a href="#about">About Us</a>
65 </nav>
66
67 <main>
68   <section id="about">
69     <h2>About the Department</h2>
70     <p>Welcome to the Computer Science Department.
71       We offer cutting-edge programs designed to prepare students for the rapidly evolving tech industry.
72       Our curriculum covers software engineering, artificial intelligence, data science, and cybersecurity.
73     </p>
74   </section>
75 </main>
76
77 <footer>
78   <p>&copy; 2026 College Name. All rights reserved.</p>
79 </footer>
80
81 <script>
82   window.onload = function() {
83     alert("Welcome to the Computer Science Department portal!");
84   };
85 </script>
86
87 </body>
88 </html>
```

Output:





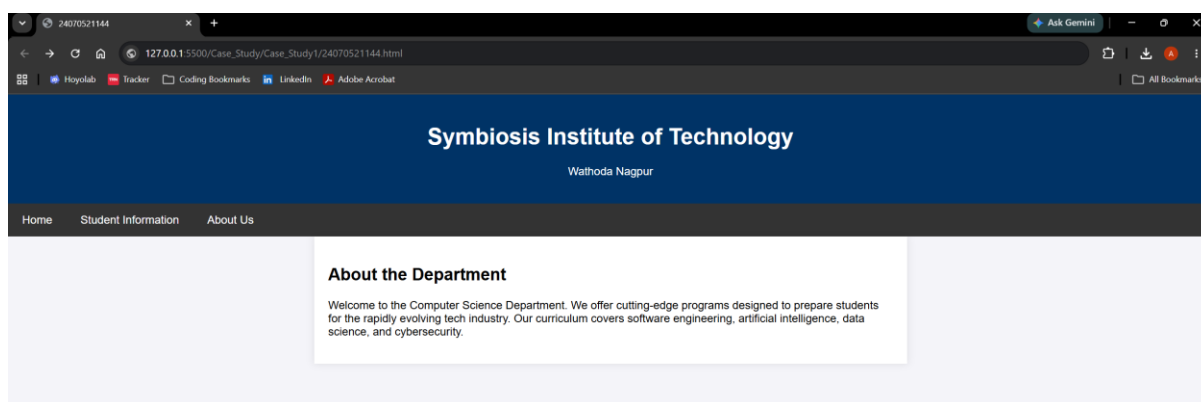
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Result: A webpage for the college department was successfully created using HTML, CSS, and JavaScript. Student information was linked and displayed correctly, and the JavaScript program successfully displayed the message **"Welcome to SIT Nagpur"** on the webpage. Thus, the aim of creating a college department webpage and implementing JavaScript functionality was achieved successfully.



Experiment: 02

Aim: Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Theory: JavaScript provides var, let, and const for declaring variables with different scopes and behaviors. It also supports modern ES6 features such as template literals for easy string formatting and object destructuring for extracting object properties into variables. The prompt() function is used to accept input from the user. In this experiment, these features are used to develop a billing calculator that accepts product details, calculates the subtotal, GST (18%), and total amount, and displays a formatted bill dynamically on the web page.

127.0.0.1:5500 says
Enter Customer Name:
User
OK Cancel

127.0.0.1:5500 says
Enter Product Name:
Coffee
OK Cancel

127.0.0.1:5500 says
Enter Product Price (₹):
249
OK Cancel

127.0.0.1:5500 says
Enter Quantity:

OK Cancel



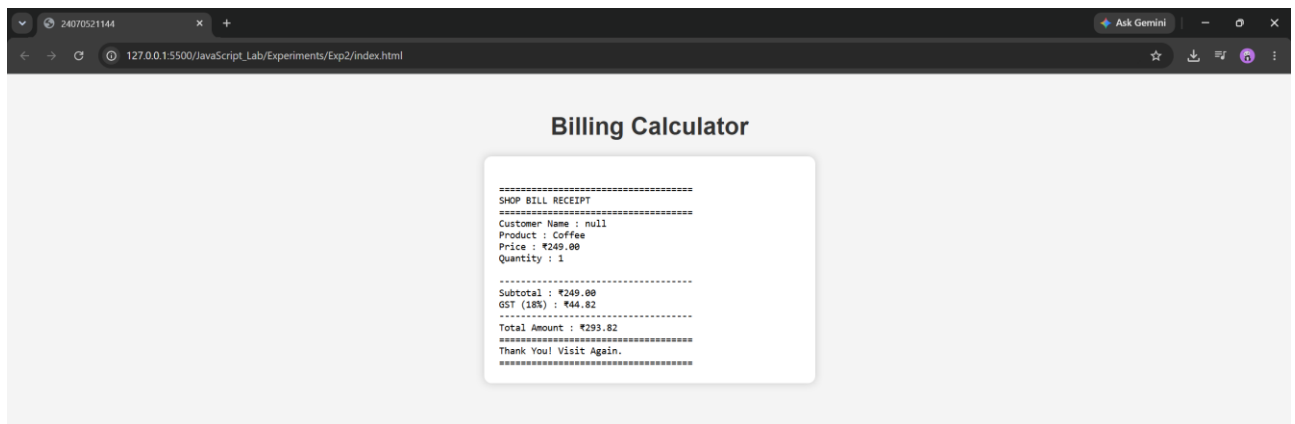
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Result : The billing calculator was successfully developed using JavaScript. It accepted user input for the customer name, product details, price, and quantity, calculated the subtotal, GST (18%), and total amount, and displayed a well-formatted billing receipt on the web page.



Case Study 02

Aim: Develop the webpage for an inline shopping website with using html, css and html tags and calculate the bill, customer name, mobile number, invoice number, product number, quantity(kg), rate per kg, discount, GST, packing charges, payment mode(cash/card/UPI), membership(yes/no)

Program Code:

```
1 <!DOCTYPE html> You, last week + Add files via upload
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>24070521144</title>
7   <!-- Link to external CSS file -->
8   <link rel="stylesheet" href="style.css">
9 </head>
10 <body>
11
12 <div class="container">
13   <h2>Online Shopping Checkout & Bill</h2>
14   <form id="billForm">
15     <div class="form-group">
16       <label for="custName">Customer Name:</label>
17       <input type="text" id="custName" required placeholder="Enter name">
18     </div>
19     <div class="form-group">
20       <label for="mobNo">Mobile Number:</label>
21       <input type="tel" id="mobNo" required placeholder="10-digit mobile number">
22     </div>
23     <div class="form-group">
24       <label for="invNo">Invoice Number:</label>
25       <input type="text" id="invNo" required value="INV-2026-001">
26     </div>
27     <div class="form-group">
28       <label for="prodName">Product Name:</label>
29       <input type="text" id="prodName" required placeholder="e.g., Organic Sugar">
30     </div>
31     <div class="form-group">
32       <label for="qtyKg">Quantity (in kg):</label>
33       <input type="number" id="qtyKg" step="0.1" min="0.1" required value="1">
34     </div>
35     <div class="form-group">
36       <label for="rateKg">Rate per kg (₹):</label>
37       <input type="number" id="rateKg" step="1" min="0" required value="100">
38     </div>
39     <div class="form-group">
40       <label for="discountPct">Discount (%):</label>
41       <input type="number" id="discountPct" step="0.1" min="0" max="100" value="5">
42     </div>
43     <div class="form-group">
44       <label for="gstPct">GST (%):</label>
45       <input type="number" id="gstPct" step="0.1" min="0" max="100" value="5">
46     </div>
47     <div class="form-group">
48       <label for="packingCharges">Packing Charges (₹):</label>
49       <input type="number" id="packingCharges" step="1" min="0" value="20">
50     </div>
51   </form>
52 </div>
```

it Graph

🕒 You, last week

🕒 AtharvaSuryas (1 week ago)



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```
45 <input type="number" id="gstPct" step="0.1" min="0" max="100" value="5">
46 </div>
47 <div class="form-group">
48 <label for="packingCharges">Packing Charges (₹):</label>
49 <input type="number" id="packingCharges" step="1" min="0" value="20">
50 </div>
51 <div class="form-group">
52 <label for="membership">Membership Type:</label>
53 <select id="membership">
54 <option value="none">None (0% extra off)</option>
55 <option value="silver">Silver (2% extra off)</option>
56 <option value="gold">Gold (5% extra off)</option>
57 </select>
58 </div>
59 <div class="form-group">
60 <label for="paymentMode">Payment Mode:</label>
61 <select id="paymentMode">
62 <option value="UPI">UPI</option>
63 <option value="Card">Credit/Debit Card</option>
64 <option value="COD">Cash on Delivery</option>
65 </select>
66 </div>
67 <button type="button" id="calcBtn">Calculate Bill</button>
68 </form>
69
70 <div class="bill-summary" id="billSummary">
71 <h3>Invoice Details</h3>
72 <div class="bill-row"><span>Customer:</span> <span id="outName"></span></div>
73 <div class="bill-row"><span>Mobile:</span> <span id="outMob"></span></div>
74 <div class="bill-row"><span>Invoice ID:</span> <span id="outInv"></span></div>
75 <div class="bill-row"><span>Product:</span> <span id="outProd"></span></div>
76 <div class="bill-row"><span>Gross Price:</span> <span id="outGross"></span></div>
77 <div class="bill-row"><span>Discount Applied:</span> <span id="outDisc"></span></div>
78 <div class="bill-row"><span>Packing Charges:</span> <span id="outPack"></span></div>
79 <div class="bill-row"><span>GST Amount:</span> <span id="outGst"></span></div>
80 <div class="bill-row"><span>Payment & Membership:</span> <span id="outMeta"></span></div>
81 ... <div class="bill-row total-row"><span>Final Payable:</span> <span id="outTotal"></span></div>
82 </div>
83 </div>
84
85 <!-- Link to external JS file -->
86 <script src="script.js"></script>
87 </body>
88 </html>
```

Git Graph

🕒 You, last week 🕒 AtharvaSuryas (1 week ago)



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Output:

Online Shopping Checkout & Bill

Customer Name:

Mobile Number:

Invoice Number:

Product Name:

Quantity (in kg):

Rate per kg (₹):

Discount (%):

GST (%):

Packing Charges (₹):

Membership Type:

Payment Mode:

Online Shopping Checkout & Bill

Customer Name:

Mobile Number:

Invoice Number:

Product Name:

Quantity (in kg):

Rate per kg (₹):

Discount (%):

GST (%):

Packing Charges (₹):

Membership Type:

Payment Mode:

Invoice Details

Customer:	User
Mobile:	1234567890
Invoice ID:	INV-2026-001
Product:	Pudina (1 kg @ ₹100/kg)
Gross Price:	₹100.00
Discount Applied:	-₹7.00 (7%)
Packing Charges:	₹20.00
GST Amount:	+₹4.65 (5%)
Payment & Membership:	COD SILVER Member
Final Payable:	₹117.65



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Result: The online shopping billing webpage was successfully developed using HTML, CSS, and JavaScript. The program accepted customer and product details, calculated the total bill including discount, GST, and packing charges, and displayed the final invoice correctly. It demonstrated the use of user input, arithmetic calculations, objects, template literals, and object destructuring in JavaScript.



Experiment No: 03

Aim: Implement control structures and form validation, create a grading system based on user entered marks

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Theory: JavaScript provides essential control structures, including conditional statements (if, else if, else, and switch) and loops (for, while, and do-while), to manage program execution based on dynamic conditions. These structures enable developers to build interactive features such as form validation, which checks and sanitizes user input on the client side before submission to prevent invalid or incomplete data. Additionally, JavaScript interacts with DOM elements to extract input values, process business logic—such as evaluating numeric ranges to assign grades—and dynamically display results on the web page. In this experiment, control structures and form validation techniques are applied to capture student marks, validate the input, dynamically calculate a grade, and display the formatted outcome to the user.

Output:

24070521144

File C:/Users/ASUS/Downloads/exp%203/index%20(3).html?id=1144

Student Grading System

Enter Marks (0 - 100):

Calculate Grade

Your Grade is: A+



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Student Grading System

Enter Marks (0 - 100):

Calculate Grade

Error: Please enter a valid number between 0 and 100.

Result:

The JavaScript-based control structures and form validation system were successfully implemented and tested. The application correctly validates user-entered marks, ensuring input fields are non-empty, numeric, and fall within the valid range of 0 to 100. Upon validation, the control logic evaluates the entered score against predefined grading criteria and dynamically displays the corresponding grade along with feedback directly on the web page.



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Case Study 03

Aim: Scholarship Eligibility, A college awards scholarships based on marks. Below 50-Not Eligible, 50-69-Bronze, 70-84-Silver, 85-94-Gold, 95-100-Platinum.

Program Code:

```
index.html ...\Case Study 3 U X  index.html ...\Case Study 2 M
Case_Study > Case Study 3 > index.html > html > title
1  <!DOCTYPE html>
2  <html>
3    <title>
4      24070521144
5    </title>
6  <head>
7    <style>
8      body{font-family:Arial,sans-serif;background: #f0f2f5;margin:0;padding:20px}
9      .card{background: #fff;padding:30px;border-radius:10px;box-shadow:0 4px 10px rgba(0,0,0,0.1);text-align:center;width:260px}
10     h2{margin-top:0;color: #333}
11     input{padding:8px;width:80%;border:1px solid #ccc;border-radius:5px;margin-bottom:12px}
12     button{padding:8px 16px;border:none;background: #4CAF50;color: #fff;border-radius:5px;cursor:pointer}
13     button:hover{background: #45a049}
14     #r{margin-top:15px;font-weight:bold;font-size:18px;color: #222}
15   </style>
16 </head>
17 <body>
18   <div class="card">
19     <h2>Scholarship Eligibility</h2>
20     Marks: <input type="number" id="m">
21     <button onclick="check()">Check</button>
22     <p id="r"></p>
23   </div>
24
25   <script>
26   function check(){
27     let m = Number(document.getElementById('m').value);
28     let r;
29     if(m<50) r="Not Eligible";
30     else if(m<70) r="Bronze";
31     else if(m<85) r="Silver";
32     else if(m<95) r="Gold";
33     else r="Platinum";
34     document.getElementById('r').innerText = "Result: " + r;
35   }
36 </script>
37 </body>
38 </html>
```



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Output:

Scholarship Eligibility

Marks:

79

Check

Result: Silver

Result: The Scholarship Eligibility was successfully implemented using JavaScript. The program validated the user input, assigned eligibility marks using if-else control structures, and displayed the correct Eligibility awards based on the entered marks. It effectively demonstrated the use of control structures and form validation in web development.



Experiment 04

Aim: Use function types, scope and closures; apply try-catch; build a palindrome checker

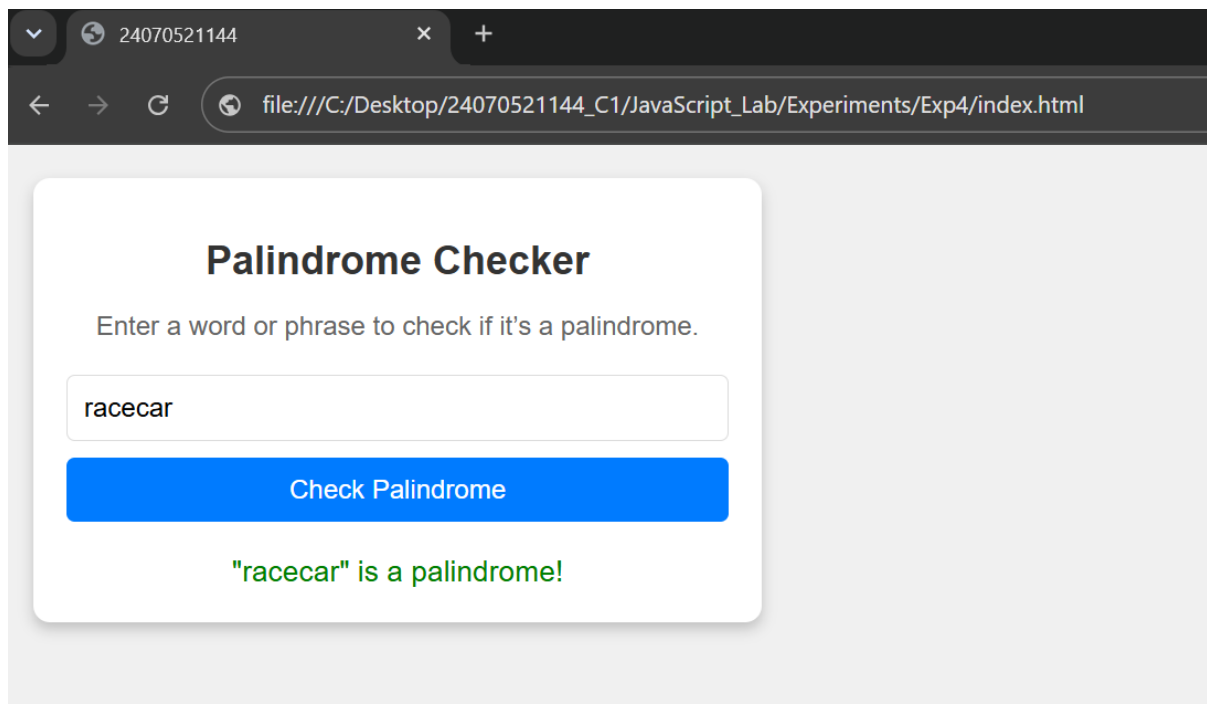
Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Theory: A palindrome is a word, number, or sentence that reads the same forward and backward, such as MADAM, LEVEL, and 121. A palindrome checker is a program that checks whether the given input is a palindrome or not by comparing the original string with its reverse.

In JavaScript, different types of functions, such as function declarations, function expressions, and arrow functions, can be used to implement the palindrome checker. Variable scope determines where variables can be accessed in the program, while closures allow an inner function to access variables from its outer function.

The try-catch statement is used for exception handling and helps manage errors, such as invalid or empty input, without stopping the execution of the program. Thus, a palindrome checker demonstrates the concepts of functions, scope, closures, and error handling in JavaScript.

Output:





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Palindrome Checker

Enter a word or phrase to check if it's a palindrome.

burger

Check Palindrome

"burger" is not a palindrome.

Result: Successfully implemented a Palindrome detector using JavaScript ES6 features, including var, let, const, template literals, and destructuring, to find and display whether the given text or number is a palindrome or not.



Case Study 04

Aim: The ATM system verifies if the pin is palindrome. If the palindrome displays a security message. To implement this create a function to reverse the entered pin number and another function to check if it is palindrome. use function declaration, function expression, arrow function and demonstrate scope and closure concepts.

Program Code:

```
Case_Study > Case Study 4 > index.html > ...
You, 2 days ago | 1 author (You)
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>PIN Strength Checker</title>
7   <style>...
58 </style>
59 </head>
60 <body>
61
62 <div class="container">
63   <h2>PIN Strength Checker</h2>
64   <p>Enter your PIN below to check its strength.</p>
65   <input type="text" id="pinInput" placeholder="Enter PIN" maxlength="8">
66   <br>
67   <button onclick="checkPinStrength()">Check Strength</button>
68   <div id="result"></div>
69 </div>
70
71 <script>
72 function checkPinStrength() {
73   const pinInput = document.getElementById('pinInput').value;
74   const resultDiv = document.getElementById('result');
75
76   // Clear previous results and check for empty input
77   resultDiv.innerHTML = '';
78   if (!pinInput) {
79     resultDiv.innerHTML = 'Please enter a PIN.';
80     resultDiv.className = '';
81     return;
82   }
83
84   // Reverse the string to check for palindrome
85   const reversedPin = pinInput.split('').reverse().join('');
86
87   if (pinInput === reversedPin) {
88     resultDiv.innerHTML = '⚠ Weak PIN (Palindrome)';
89     resultDiv.className = 'weak';
90   } else {
91     resultDiv.innerHTML = '✅ Strong PIN';
92     resultDiv.className = 'strong';
93   }
94 }
95 </script>
96
97 </body>
98 </html>
```



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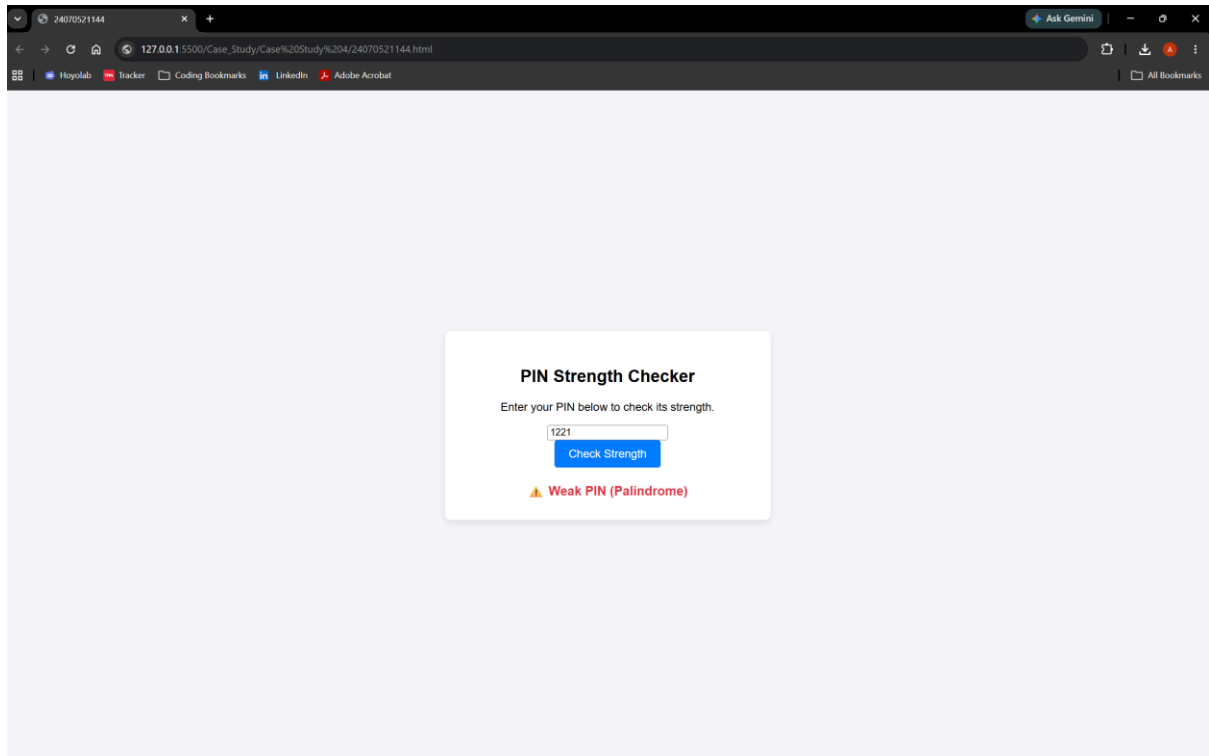
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Output:



Result: Successfully implemented a ATM pin Palindrome detector using JavaScript ES6 features, including var, let, const, template literals, and destructuring, to find and display whether the given ATM pin is a palindrome or not.